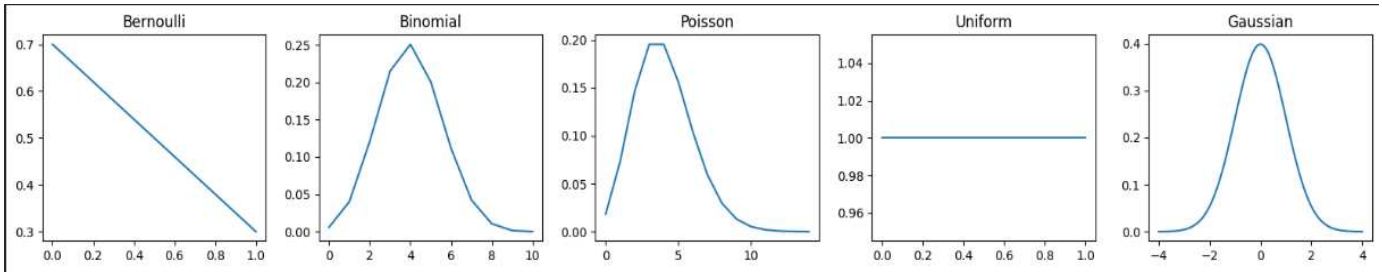
EXPERIMENT3: Probability Distributions and Bayes' Theorem
Problem Statement: A diagnostic centre uses a screening test for a disease. Doctors need the probability that a patient with a positive result actually has the disease. Disease prevalence is 1%, test
sensitivity 95% and specificity 90%. Patient age and test-score data follow known distributions. The

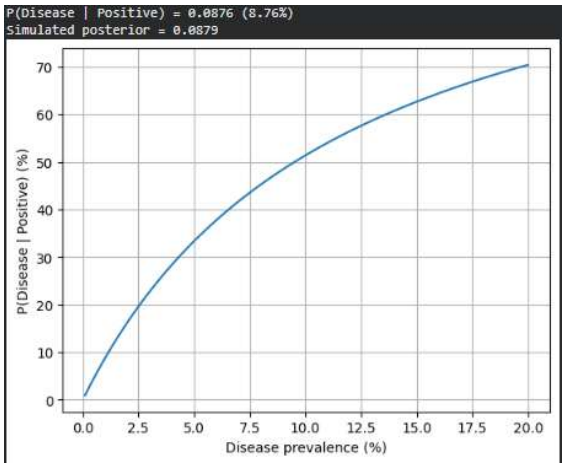
objective is to model data with standard distributions and apply Bayes' theorem to a real decision. Use
NumPy, SciPy, Matplotlib.
Tasks:
1. Generate and plot Bernoulli, Binomial, Poisson, Uniform and Gaussian samples with SciPy; compare with their PDFs/PMFs.
2. Compute P(Disease | Positive) using Bayes' theorem.
3. Simulate 100,000 patients and verify the result empirically.
4. Plot how the posterior changes as prevalence varies from 0.1% to 20%.

AIM:
To generate and visualise different probability distributions using SciPy and apply Bayes' theorem to calculate the probability of disease given a positive test result.
PROGRAM:

```
from scipy.stats import bernoulli, binom, poisson, uniform, norm
fig,axs=plt.subplots(1,5,figsize=(16,3))
dists=[('Bernoulli',bernoulli(.3),[0,1]),('Binomial',binom(10,.4),np.arang
e(11)),('Poisson',poisson(4),np.arange(15)),('Uniform',uniform(0,1),np.lin
space(0,1,100)),('Gaussian',norm(0,1),np.linspace(-4,4,100))]
for ax,(name,d,x) in zip(axs,dists):
 y=d.pmf(x) if name in ['Bernoulli','Binomial','Poisson'] else d.pdf(x)
 ax.plot(x,y); ax.set_title(name)
plt.tight_layout(); plt.show()
prev=.01; sens=.95; spec=.90
post=sens*prev/(sens*prev+(1-spec)*(1-prev))
print("P(Disease | Positive) =",round(post,4),f"({post*100:.2f}%)")
N=100000; disease=np.random.rand(N)<prev
positive=np.where(disease,np.random.rand(N)<sens,np.random.rand(N)<(1-spec
))
print("Simulated posterior =",round(disease[positive].mean(),4))
p=np.linspace(.001,.20,200); posterior=sens*p/(sens*p+(1-spec)*(1-p))
plt.plot(p*100,posterior*100); plt.xlabel("Disease prevalence (%)");
plt.ylabel("P(Disease | Positive) (%)"); plt.grid(); plt.show()
```

OUTPUT:





RESULT:

Bernoulli, Binomial, Poisson, Uniform and Gaussian distributions were generated and plotted.
The posterior probability was calculated using Bayes' theorem and verified through simulation.